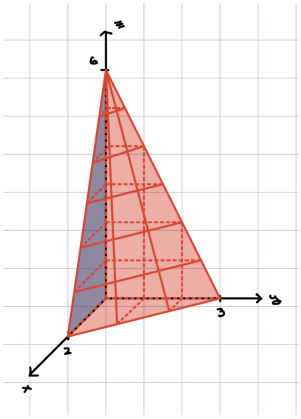
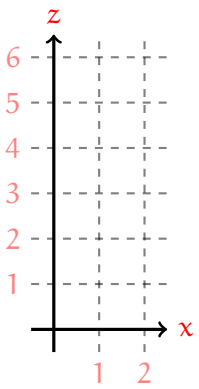
**Example 1.** Find the volume of the tetrahedron T bounded by the planes: $3x + 2y + z = 6$, $x = 0$, $y = 0$, and $z = 0$, but in the order: $dydzdx$. Last time, we did this in a different order of integration, and found answer 6.



**Solution.** The outer integration variables are $dzdx$. So we sketch the $xz$–shadow.



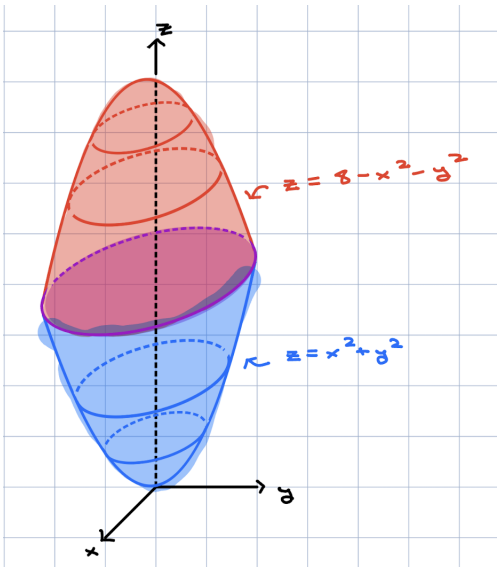

For each fixed $x$ in:

the range of $z$ in the $xz$–shadow is:

For each fixed $(x, z)$ in the $xz$–shadow, the range of $y$ is:

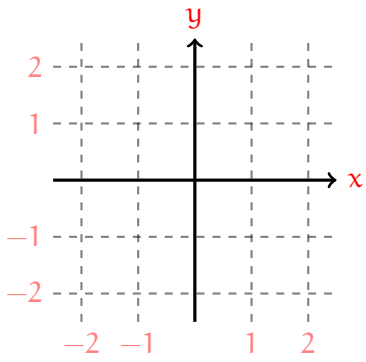
[volume of T] =

**Example 2.** If $E$ is the region bounded by the paraboloids $z = 8 - x^2 - y^2$ and $z = x^2 + y^2$ then set up an iterated integral of an arbitrary function $f(x, y, z)$ over $E$ in the in order $dz\,dy\,dx$.




**Solution.** The outer integration variables are $dy\,dx$. So we sketch the $xy$–shadow.

The edge of the $xy$–shadow is the edge of the thickest part of $E$. This occurs when




For each fixed $x$ in:

the range of $y$ in the $xy$–shadow is:

For each fixed $(x, y)$ in the $xy$–shadow, the range of $z$ is:

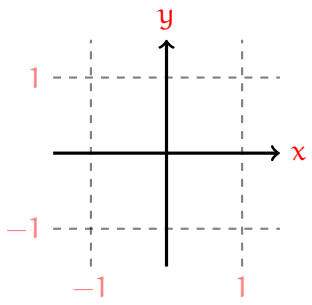
Thus: $\iiint_E f(x, y, z)\ dV =$

**Example 3.** Sketch the region of integration for:

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{x^2+y^2}} f(x,y,z)\, dz dy dx$$

**Solution.** The outer integration variables are $dydx$.

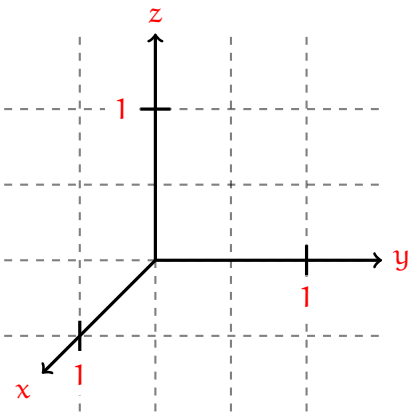
Thus the outer bounds describe the $xy$–shadow.




The inner integration variable is $dz$.

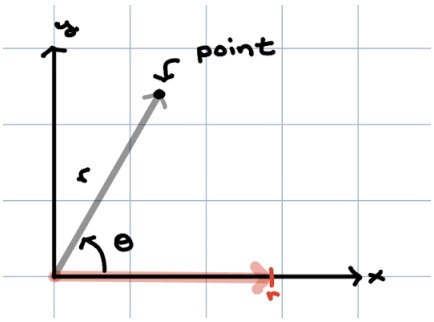
Thus the inner bounds describe the top and bottom of our region.

We sketch the vertical space between top and bottom, for $(x,y)$ in the $xy$–shadow.




We can also view the region in Desmos.

A. **Polar Coordinates.** **Polar coordinates** $r$ and $\theta$ are described as followed.




when $r$ is positive

- $r$ measures:
- $\theta$ measures:

when $r$ is flipped negative the position vector of the point is also:

Sketch the points with polar coordinates $[r = 2, \theta = \frac{\pi}{2}]$ and $[r = -2, \theta = \frac{3\pi}{2}]$.

